



Savitribai Phule Pune University
Centre For Distance and Online Education

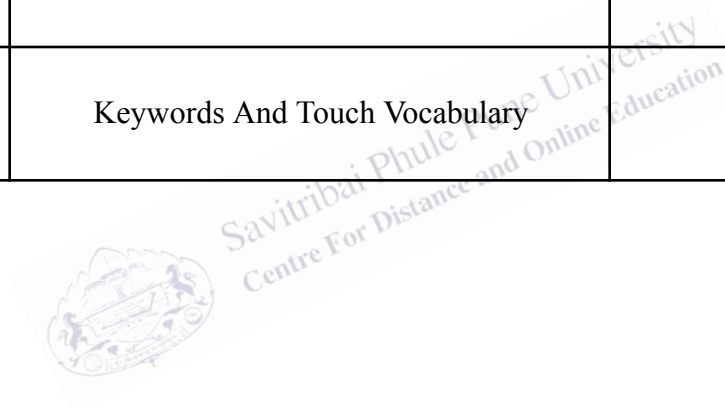
SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

CENTRE FOR DISTANCE AND ONLINE EDUCATION

Program	Master of Computer Application		
Course	Programming from First Principles		
Topic Name	Practical Session Part - 4 Combination		
Topic Code	-		
Unit/Module No. & Name	12		Recursion
Self-Learning Hours	-		

Topic Details

S.No	Topic	Pg.No
1.0	Introduction	4
2.0	Meaning And Definition	5-8
3.0	Nature Or Type	9-12
4.0	Examples And Case Studies	13-17
5.0	Keywords And Touch Vocabulary	18



OBJECTIVE

1. This unit explains the idea of combination in a clear and simple way. The main goal is to help undergraduate students understand how to choose objects from a group when order does not matter.
2. Students often meet problems in mathematics, computer science, economics, business, and data analysis where they must count possible selections.
3. A good understanding of combination builds strong logical thinking and careful problem-solving habits.
4. After studying this unit, students should be able to define combination, identify its main features, use the standard formula, and solve basic real-life problems.
5. Students should also learn how to decide whether a question belongs to combination or another counting method, such as permutation. This distinction is very important because many errors happen when students count arrangements instead of choices.
6. Combination is a foundation topic in discrete mathematics and quantitative reasoning. It supports later topics such as probability, binomial expansion, sampling, coding, algorithm design, and research methods.
7. When students know how combinations work, they can read data problems more carefully and write better logical answers.
8. Combination is not only a classroom topic. People use this idea while forming teams, selecting menu items, choosing subjects, making product bundles, or planning surveys.
9. The method teaches us to think about all possible choices in a structured way.

1.0 INTRODUCTION

Mathematics is not only about numbers and formulas. It is also about patterns, choices, and logical systems. In many situations, we ask, “How many ways can this be done?” Counting methods help answer that question. Combination is one of the most useful counting methods because it deals with selections where position has no importance.

Suppose a teacher must choose three class representatives from 10 students. Suppose a shop owner wants to design a snack pack with four items from a list of 12. Suppose a researcher wants to select five people from a group for interviews. In all these cases, the group chosen matters, but the order of names does not change the answer. A team of A, B, and C is the same as a team of C, A, and B. This simple idea is the heart of combination.

Students sometimes feel that counting questions are easy because the numbers may look small. However, counting requires deep care. A small change in wording can change the method. If the problem asks for an arrangement, order matters. If the problem asks for a selection, order does not matter. Good readers notice this difference before applying any formula.

Learning combination also develops mathematical maturity. This means students learn how to read a problem, identify hidden conditions, break it into parts, and choose a correct method. These are useful academic skills in every subject. Even when students forget a formula, they can still reason through small examples and rebuild the logic.

Counting ideas have grown over many centuries. Early scholars studied games, trade, astronomy, and patterns. Over time, counting methods became part of probability and later part of modern statistics and computer science. Today, combination supports fields that use data, models, networks, and decision systems.

In present-day education, combination matters because modern life produces many choices. Students compare products, researchers design samples, and programmers test cases. In each area, a correct count helps people make sound conclusions. Therefore, this topic is both old in history and modern in use.

2.0 MEANING AND DEFINITION

2.1 Meaning Of Combination

2.1.1 Simple Meaning

A combination is a way of choosing some objects from a larger group without caring about order. The focus is on membership in the group, not on position. If we choose two fruits, apple and mango, that choice is the same as mango and apple. The set of selected items remains the same, so it counts as one combination, not two.

2.1.2 Everyday Meaning

In common life, people use the word combination for a mix or set. In mathematics, the word has a more exact meaning. It refers to a counted selection from available items under clear rules. Because the word has a formal meaning here, students must read it as a technical term, not as casual speech.

2.2 Formal Definition

2.2.1 Standard Definition

A combination of r objects from a set of n distinct objects is a selection of r objects where the order of selection is not important. The number of such combinations is written as nCr or $C(n, r)$. This notation tells us how many different groups of size r can be formed from n different objects.

2.2.2 Formula

The standard formula is:

$$nCr = n! / [r! (n - r)!]$$

In this formula, the symbol $!$ means factorial. Factorial is the product of positive whole numbers from 1 up to that number. For example, $4!$ means $4 \times 3 \times 2 \times 1$, which equals 24. The formula removes repeated counting that would happen if we counted every order of the same group.

2.3 Why Order Does Not Matter

2.3.1 Difference In Logic

Order matters in a race, a password, a seating plan, or a rank list. Order does not matter in a team, a committee, a basket of chosen fruits, or a set of books borrowed together. This difference changes the answer. If students ignore order, they may count too few or too many results.

2.3.2 Link With Sets

Combination is closely linked with the mathematical idea of a set. In a set, the order of elements is not important. The set $\{1, 2, 3\}$ is the same as $\{3, 2, 1\}$. This is why combination works well when a question is about forming groups rather than arranging positions.

2.4 Relation With Permutation

2.4.1 Key Difference

Permutation and combination are related, but they are not the same. Permutation counts arrangements where order matters. Combination counts selections where order does not matter. For example, the ordered pairs AB and BA are two permutations, but they form only one combination when the task is simply to choose two letters.

2.4.2 Practical Test

A useful test is to ask, “If I switch the order, do I get a new answer?” If yes, the problem is about permutation. If no, the problem is about combination. This simple test helps students avoid confusion.

2.5 Main Properties

2.5.1 Boundary Values

Some values of combination are easy to understand. If we choose zero objects from n objects, there is only one way: choose nothing. So, $nC_0 = 1$. If we choose all n objects from n objects, there is also only one way: choose the full set. So, $nC_n = 1$.

2.5.2 Symmetry Rule

Combination has an important symmetry rule: $nCr = nC(n - r)$. This means choosing r objects is equal in number to leaving behind n minus r objects. For example, choosing two students out of six gives the same count as leaving out four students from six. Both counts are equal because each chosen group matches one unchosen group.

Fig. 1

Understanding Combinations



This visual explains combinations through simple questions and answers, defining them as selections without order, contrasting them with permutations, and emphasizing the importance of grouping over arrangement.

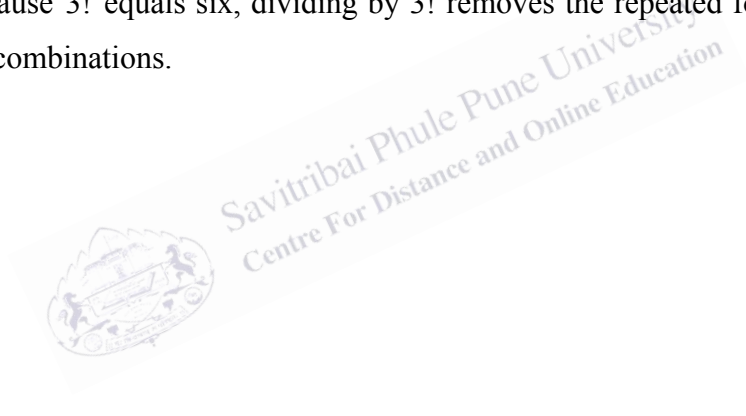
2.6 Understanding The Formula

2.6.1 Why Factorial Appears

Many students ask why factorial appears in the formula for combination. The reason comes from repeated counting. If we first count every ordered arrangement of r objects from n objects, then each selected group is counted many times. For a group of r objects, the same members can be arranged in $r!$ ways. Since all those arrangements represent one true selection, we divide by $r!$ to remove extra counting. This is why combination is smaller than permutation.

2.6.2 A Small Illustration

Imagine we choose three letters from A, B, C, and D. The group A, B, and C can appear as ABC, ACB, BAC, BCA, CAB, and CBA. These six forms are different arrangements, but they are only one selection. Because $3!$ equals six, dividing by $3!$ removes the repeated forms and leaves the correct number of combinations.



3.0 NATURE OR TYPE

3.1 Nature Of Combination

3.1.1 Selection-Based Nature

The first nature of combination is that it is selection-based. It asks who or what is included in a group. It does not ask where each item stands. This makes combination suitable for committees, teams, bundles, samples, and many similar tasks.

3.1.2 Distinct-Object Nature

The usual form of combination starts with distinct objects. Distinct means each item can be told apart from the others. The items may be people, books, cards, numbers, or products. When items are distinct, each chosen group can be identified clearly.

3.2 Mathematical Nature

3.2.1 Finite Counting Nature

Combination works within a finite set. That means the number of available objects is limited and countable. The method is exact, not approximate. It gives a fixed answer based on the given conditions. This exactness is one reason why it is useful in academic work.

3.2.2 Rule-Governed Nature

Combination is controlled by rules. Students must know the number of available objects, the number to be selected, and any conditions such as required members or forbidden members. A correct answer depends on following all conditions carefully.

3.3 Types Of Combination

3.3.1 Combination Without Repetition

This is the most common type. In this type, each object can be chosen only once. For example, if a student selects three different books from a shelf, the same book cannot be chosen again in the same selection. The formula nCr is used for this type.

3.3.2 Combination With Repetition

In some situations, the same type of item may be chosen more than once. For example, a customer may choose three ice cream scoops from chocolate, vanilla, and strawberry, and the same flavor can be repeated. This type is called combination with repetition. Its logic is different from the simple nCr formula. Students should first identify whether repeated choice is allowed.

3.4 Restricted Combination

3.4.1 With Required Elements

Sometimes a question says that one or more items must be included. For example, a committee of four must include the class monitor. In such a case, the required person is fixed first, and the remaining members are chosen from the remaining people.

3.4.2 With Excluded Elements

Some questions say that certain items cannot be included. For example, a team of five must be formed from 12 players, but two injured players cannot be selected. In that case, the available group becomes smaller, and the count is taken from the reduced set.

3.5 Comparative Nature

3.5.1 Efficient Counting

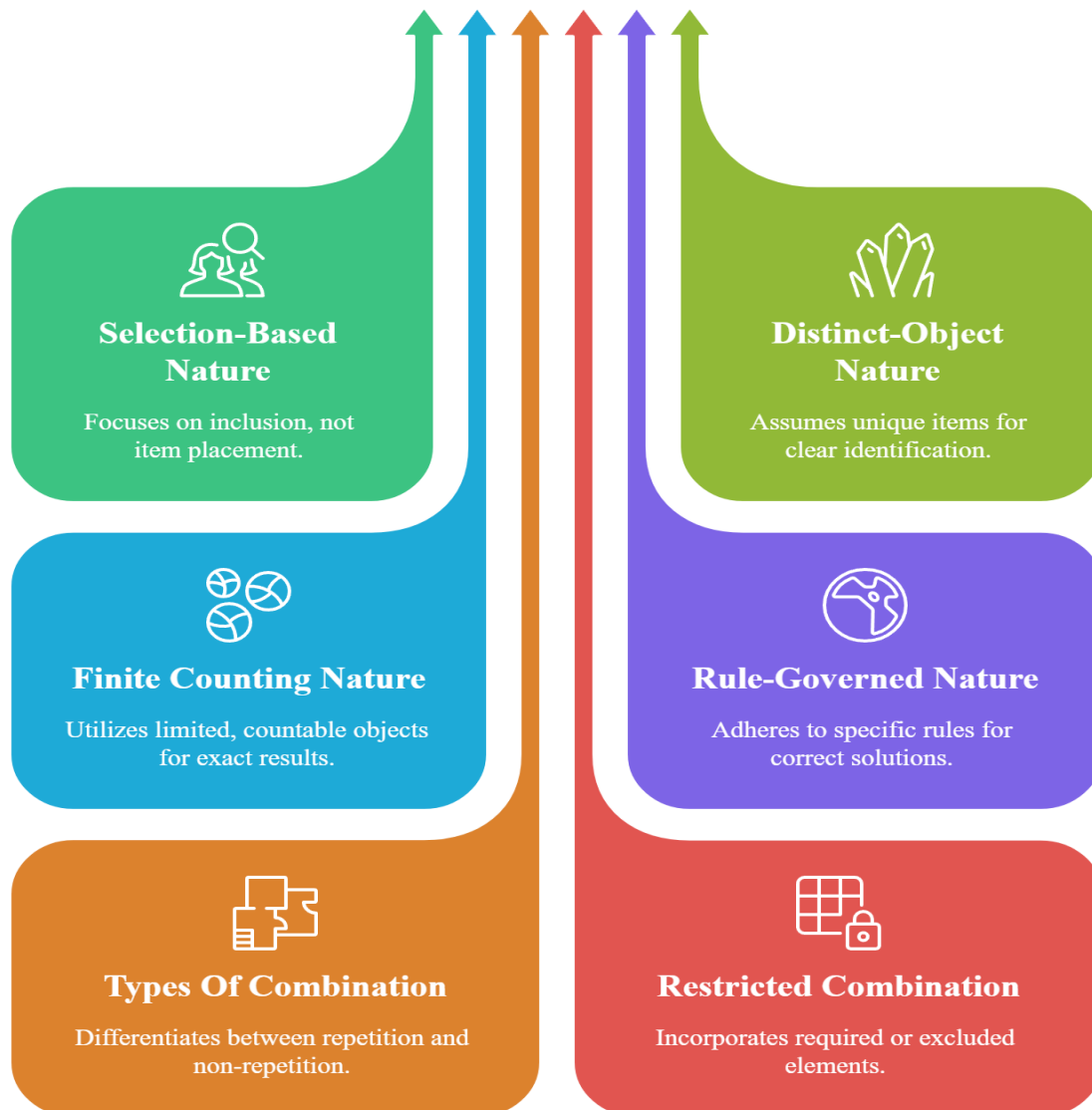
Combination is efficient because it avoids listing every possible group one by one. When numbers are large, listing is slow and risky. The formula gives a shorter and safer method.

3.5.2 Logical Economy

It also shows logical economy. Many arrangements may represent the same selected group. Combination removes such repeated counting and gives one count for one true group. This makes the method neat and academically powerful.

Fig. 2

Pathways to Combination Accuracy



This infographic illustrates key pathways to mastering combinations, highlighting selection-based thinking, distinct objects, finite counting, rule adherence, types of combinations, and handling restrictions for accurate problem-solving.

3.6 Interpretive Types In Practice

3.6.1 Small-Number Listing Type

When the numbers are small, students can list all possible groups by hand. This type is useful for learning because it shows the real meaning of a combination. For example, choosing two letters from A, B, C, and D can be listed as AB, AC, AD, BC, BD, and CD. Listing helps beginners see that BA is not a new result.

3.6.2 Large-Number Formula Type

When the numbers become large, listing is no longer practical. In such cases, the formula method is better. For example, choosing six students from 20 students would take too long to list by hand. The formula gives the answer quickly and reduces human error.

3.7 Conceptual Features

3.7.1 Independence From Position

Another key feature of combination is independence from position. The chosen members do not hold numbered places inside the result. This is why combination is not suitable for ranking, seating, or code creation. Those tasks depend on exact placement and must use a different method.

3.7.2 Dependence On Conditions

At the same time, combination strongly depends on stated conditions. Even a short phrase such as “at least one,” “exactly two,” or “no repetition” changes the solution path. Good students do not rush to the formula. They first translate the language of the question into a mathematical plan.

4.0 EXAMPLES AND CASE STUDIES

4.1 Worked Examples

4.1.1 Choosing Students

A teacher wants to choose two students from a group of five students: A, B, C, D, and E. The possible groups are AB, AC, AD, AE, BC, BD, BE, CD, CE, and DE. There are 10 groups in total. By formula, ${}^5C_2 = 5! / (2!3!) = 10$. The listed answer and the formula answer match. This example shows how the formula saves time after the idea becomes clear.

4.1.2 Choosing Books

A student wants to borrow three books from a shelf of seven different books. The order of borrowing does not matter because the student takes the same group of books even if the books are picked in a different order. So the answer is ${}^7C_3 = 35$. This means 35 different groups of three books can be borrowed.

4.2 Applied Examples

4.2.1 Forming A Committee

Suppose a college club has 12 members and needs a committee of four. The number of possible committees is ${}^{12}C_4 = 495$. This value tells us how many different groups of four can be formed. It does not tell us the order of speaking, seating, or rank. It only counts the group itself.

4.2.2 Selecting Survey Participants

A researcher wants to choose five participants from 15 volunteers for an interview study. Because only the selected group matters, the answer is ${}^{15}C_5 = 3,003$. This shows how quickly the number of possibilities grows when the larger group increases.

4.3 Case Study One

4.3.1 School Festival Team

A college is planning a school festival. Ten students volunteer to manage the event, but only three can join the core planning team. The teacher wants to know how many different teams are possible. Since the three selected students form one team no matter how their names are ordered, the method is combination. Therefore, the number of possible teams is ${}^{10}C_3 = 120$.

4.3.2 Academic Insight

This case study shows why combination matters in administration and management. Selection problems appear in student councils, project teams, interview panels, and club leadership groups. A correct count helps ensure fairness, transparency, and planning.

4.4 Case Study Two

4.4.1 Product Bundle Design

A small food business wants to create gift packs. The owner has eight snack items and wants each gift pack to contain three different items. The owner needs to know how many distinct gift packs can be offered to customers. Because the order of items inside the pack does not create a new pack, the problem is a combination problem. The number of packs is ${}^8C_3 = 56$.

4.4.2 Business Value

This example shows that combination is useful in business decisions. It helps estimate variety, manage stock plans, and design product options. A business can use the count to decide whether its menu or product line is too narrow or too wide.

4.5 Case Study Three

4.5.1 Basic Probability Link

A bag contains six different colored balls. A student picks two balls at random. Before studying probability, the student must know how many different pairs are possible. Since the order of picking does not matter for the pair itself, the total number of possible pairs is ${}^6C_2 = 15$. This total becomes the sample space for later probability work.

4.5.2 Research Value

This case study shows how combination supports later topics. In probability, statistics, and data science, students often need the total number of possible samples or outcomes. Combination helps build that total correctly.

4.6 Common Errors And Corrections

4.6.1 Confusing Combination With Permutation

One common mistake is using permutation when the question is only about selection. For instance, a student may count AB and BA separately when choosing two classmates for a team. This gives a larger answer than the correct one. The correction is to ask whether order creates a new result.

4.6.2 Ignoring Conditions

Another mistake is forgetting conditions such as “must include one girl,” “must exclude one absent student,” or “repetition allowed.” These words change the method. Students should underline key conditions before starting the solution.

4.7 Stepwise Method For Solving Problems

4.7.1 Read And Identify

First, read the question slowly. Find the total number of objects, the number to be selected, and any special rule. Ask whether the problem is about selecting or arranging.

4.7.2 Choose The Method

Second, test whether order matters. If it does not matter, use combination. If repetition is not allowed, use nCr . If repetition is allowed, use the method for repeated selections.

4.7.3 Compute Carefully

Third, substitute the values into the formula and simplify step by step. Students should reduce common factors when possible to avoid large calculations. This makes the work shorter and safer.

4.7.4 Check The Answer

Finally, check whether the answer makes sense. The answer should be a whole number. It should not be larger than the number of possible arrangements when order matters. For small problems, students may list the groups to verify the formula.

1. Combination counts selection, not arrangement.
2. The formula nCr is used when order does not matter and repetition is not allowed.
3. Careful reading of conditions prevents many mistakes.
4. Combination supports probability, statistics, business, and computing.
5. A quick test is to switch the order and see whether the answer changes.

4.8 Extended Worked Example

4.8.1 Sports Team Selection

A college has nine players available, and the coach must choose four players for a practice match. Since the coach is only selecting the group, order does not matter. Therefore, the answer is 9C_4 . Using the formula, ${}^9C_4 = 9! / (4!5!) = 126$. This means 126 different groups of four players can be chosen.

4.8.2 With A Required Player

Now suppose the captain must be included in the team of four. Then one place is already filled by the captain. The coach only needs to choose three more players from the remaining eight players. So the number of possible teams becomes ${}^8C_3 = 56$. This example shows how a required condition changes the count.

4.9 Extended Case Study

4.9.1 Library Display Planning

A college library wants to create a monthly display of recommended books. There are 14 new books, but the display shelf has space for only five books. The librarian wants to know how

many different sets of five books can be shown over time. Because the display is about the chosen set and not about the order in which books are placed for this counting task, the correct method is combination. The answer is ${}^{14}C_5 = 2,002$.

4.9.2 Administrative Use

This case study shows how combination helps in educational management. Libraries, departments, and event offices often work with limited spaces and many options. Combination helps them estimate variety and plan rotation schedules in a rational way.

4.10 Link With Probability And Research

4.10.1 Sampling Logic

In research, a sample is a smaller group taken from a larger population. When a researcher chooses a sample without caring about order, combination gives the number of possible samples. This matters in survey design, fieldwork planning, and random selection. Even when software performs the choice, understanding the counting logic helps the researcher explain the process clearly.

4.10.2 Fairness And Transparency

Combination also supports fairness. If every group of the same size is counted equally, then the selection process can be explained in a transparent way. This is important in academic studies, student committees, and institutional tasks where trust matters.

5.0 Keyword Table

Keyword	Simple Academic Meaning
Combination	A selection of items where order does not matter
Selection	The act of choosing items from a group
Order	The position or arrangement of chosen items
Distinct	Different and separately identifiable
Factorial	The product of whole numbers down to one
Formula	A rule written with symbols for calculation
Permutation	An arrangement where order matters
Repetition	Choosing the same item more than once when allowed
Committee	A selected group formed for a task
Sample Space	The full set of possible outcomes
Restriction	A condition that limits the choice
Symmetry	A property that shows two counts are equal